R Project 01

2024-11-22

R Markdown

This is an R Markdown document. Markdown is a simple formatting syntax for authoring HTML, PDF, and MS Word documents. For more details on using R Markdown see http://rmarkdown.rstudio.com.

When you click the **Knit** button a document will be generated that includes both content as well as the output of any embedded R code chunks within the document. You can embed an R code chunk like this:

```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
r_project_data <- read.csv("/Users/chasegolden/Documents/r project data.csv")

head(r_project_data)
```

```
##   X work_year experience_level employment_type                  job_title
## 1 0      2020               MI              FT             Data Scientist
## 2 1      2020               SE              FT Machine Learning Scientist
## 3 2      2020               SE              FT          Big Data Engineer
## 4 3      2020               MI              FT       Product Data Analyst
## 5 4      2020               SE              FT  Machine Learning Engineer
## 6 5      2020               EN              FT               Data Analyst
##   salary salary_currency salary_in_usd employee_residence remote_ratio
## 1  70000             EUR         79833                 DE            0
## 2 260000             USD        260000                 JP            0
## 3  85000             GBP        109024                 GB           50
## 4  20000             USD         20000                 HN            0
## 5 150000             USD        150000                 US           50
## 6  72000             USD         72000                 US          100
##   company_location company_size
## 1               DE            L
## 2               JP            S
## 3               GB            M
## 4               HN            S
## 5               US            L
## 6               US            L
```

```
getwd()
```

```
## [1] "/Users/chasegolden"
```

```r
head(r_project_data)
```

```
##   X work_year experience_level employment_type                  job_title
## 1 0      2020               MI              FT             Data Scientist
## 2 1      2020               SE              FT Machine Learning Scientist
## 3 2      2020               SE              FT          Big Data Engineer
## 4 3      2020               MI              FT       Product Data Analyst
## 5 4      2020               SE              FT  Machine Learning Engineer
## 6 5      2020               EN              FT               Data Analyst
##   salary salary_currency salary_in_usd employee_residence remote_ratio
## 1  70000             EUR         79833                 DE            0
## 2 260000             USD        260000                 JP            0
## 3  85000             GBP        109024                 GB           50
## 4  20000             USD         20000                 HN            0
## 5 150000             USD        150000                 US           50
## 6  72000             USD         72000                 US          100
##   company_location company_size
## 1               DE            L
## 2               JP            S
## 3               GB            M
## 4               HN            S
## 5               US            L
## 6               US            L
```

```r
library(readr)
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

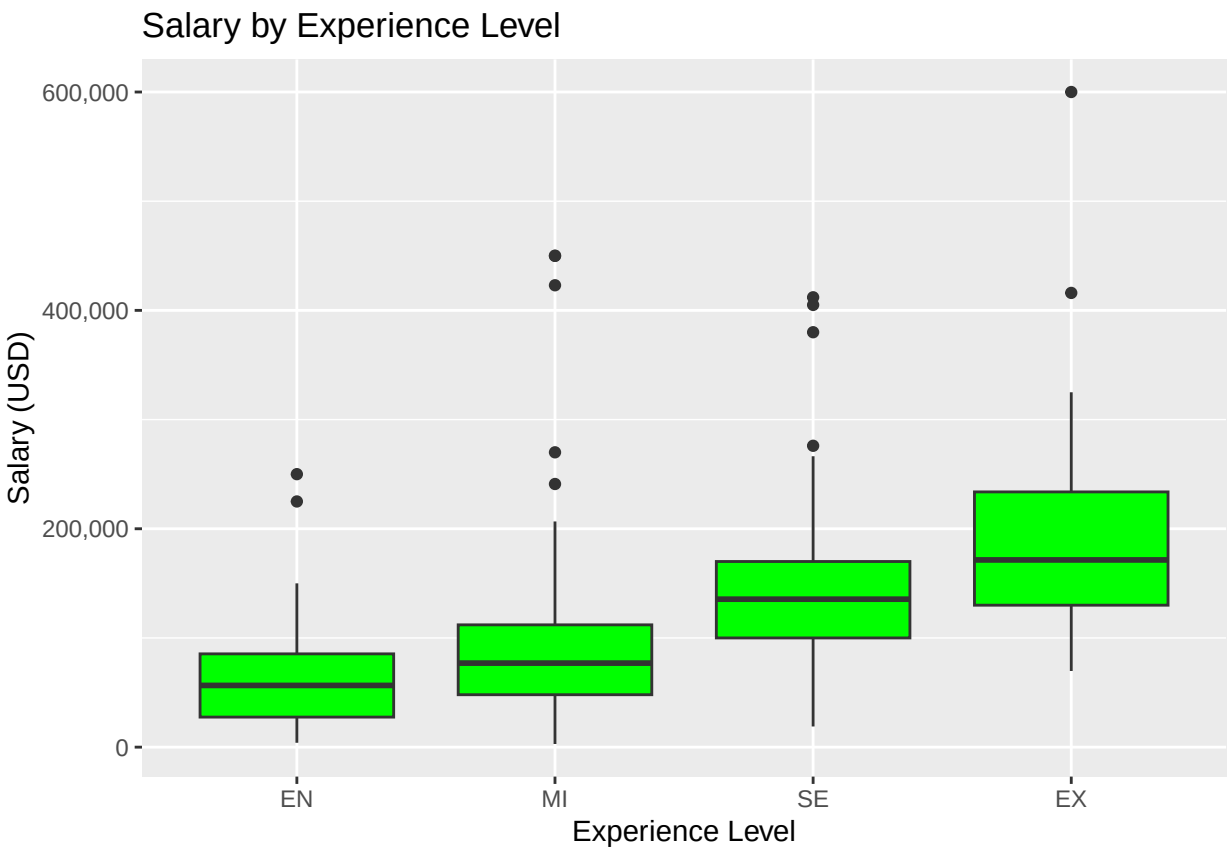
```r
library(lubridate)
library(ggplot2)
library(dplyr)
library(scales)
```

```
##
## Attaching package: 'scales'
##
## The following object is masked from 'package:purrr':
##
##     discard
##
## The following object is masked from 'package:readr':
##
##     col_factor
```

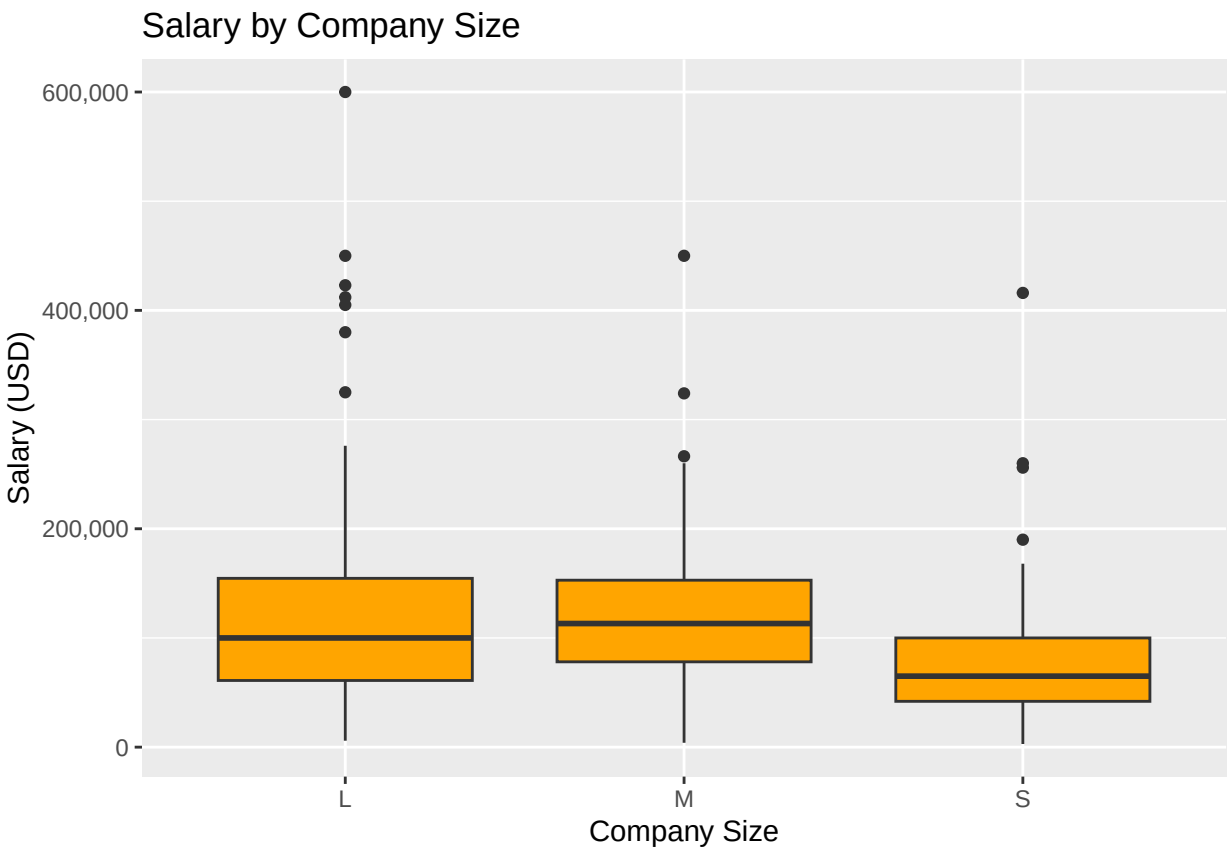
```r
r_project_data_1 <- r_project_data %>%
  mutate(
    work_year = as.integer(work_year),
    salary_in_usd = as.numeric(salary_in_usd),
    experience_level = factor(experience_level, levels = c("EN", "MI", "SE", "EX")),
    employment_type = factor(employment_type),
    company_size = factor(company_size),
    job_title = factor(job_title)
  )

summary(r_project_data_1)
```

```
##        X           work_year    experience_level employment_type
##  Min.   :  0.0   Min.   :2020   EN: 88           CT:  5
##  1st Qu.:151.5   1st Qu.:2021   MI:213           FL:  4
##  Median :303.0   Median :2022   SE:280           FT:588
##  Mean   :303.0   Mean   :2021   EX: 26           PT: 10
##  3rd Qu.:454.5   3rd Qu.:2022
##  Max.   :606.0   Max.   :2022
##
##                      job_title       salary         salary_currency
##  Data Scientist           :143   Min.   :    4000   Length:607
##  Data Engineer            :132   1st Qu.:   70000   Class :character
##  Data Analyst             : 97   Median :  115000   Mode  :character
##  Machine Learning Engineer: 41   Mean   :  324000
##  Research Scientist       : 16   3rd Qu.:  165000
##  Data Science Manager     : 12   Max.   :30400000
##  (Other)                  :166
##  salary_in_usd    employee_residence  remote_ratio    company_location
##  Min.   :  2859   Length:607         Min.   :  0.00   Length:607
##  1st Qu.: 62726   Class :character   1st Qu.: 50.00   Class :character
##  Median :101570   Mode  :character   Median :100.00   Mode  :character
##  Mean   :112298                      Mean   : 70.92
##  3rd Qu.:150000                      3rd Qu.:100.00
##  Max.   :600000                      Max.   :100.00
##
##  company_size
##  L:198
##  M:326
##  S: 83
##
##
##
##
```

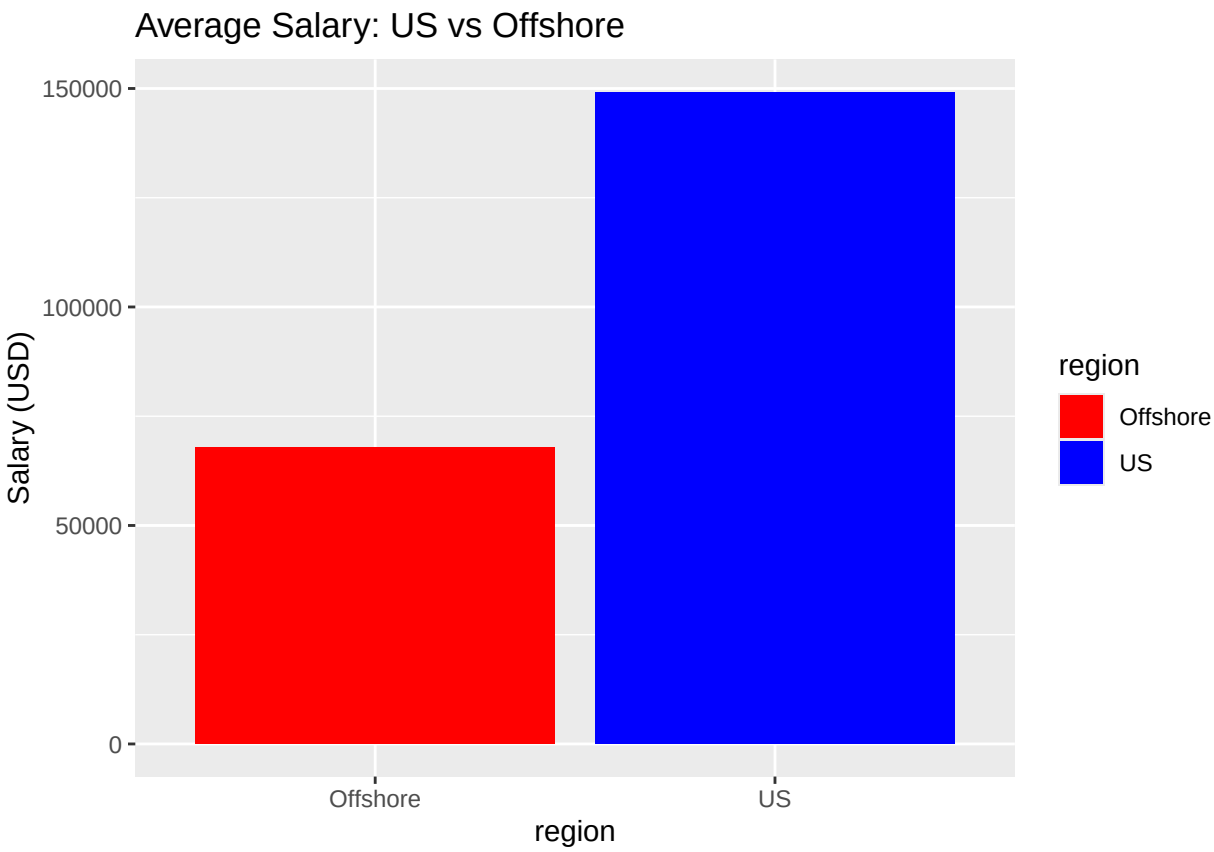
```r
#Salary by Experience Level
ggplot(r_project_data_1, aes(x = experience_level, y = salary_in_usd)) +
  geom_boxplot(fill = "green") +
  labs(title = "Salary by Experience Level", y = "Salary (USD)", x = "Experience Level") +
  scale_y_continuous(labels = label_comma())
```



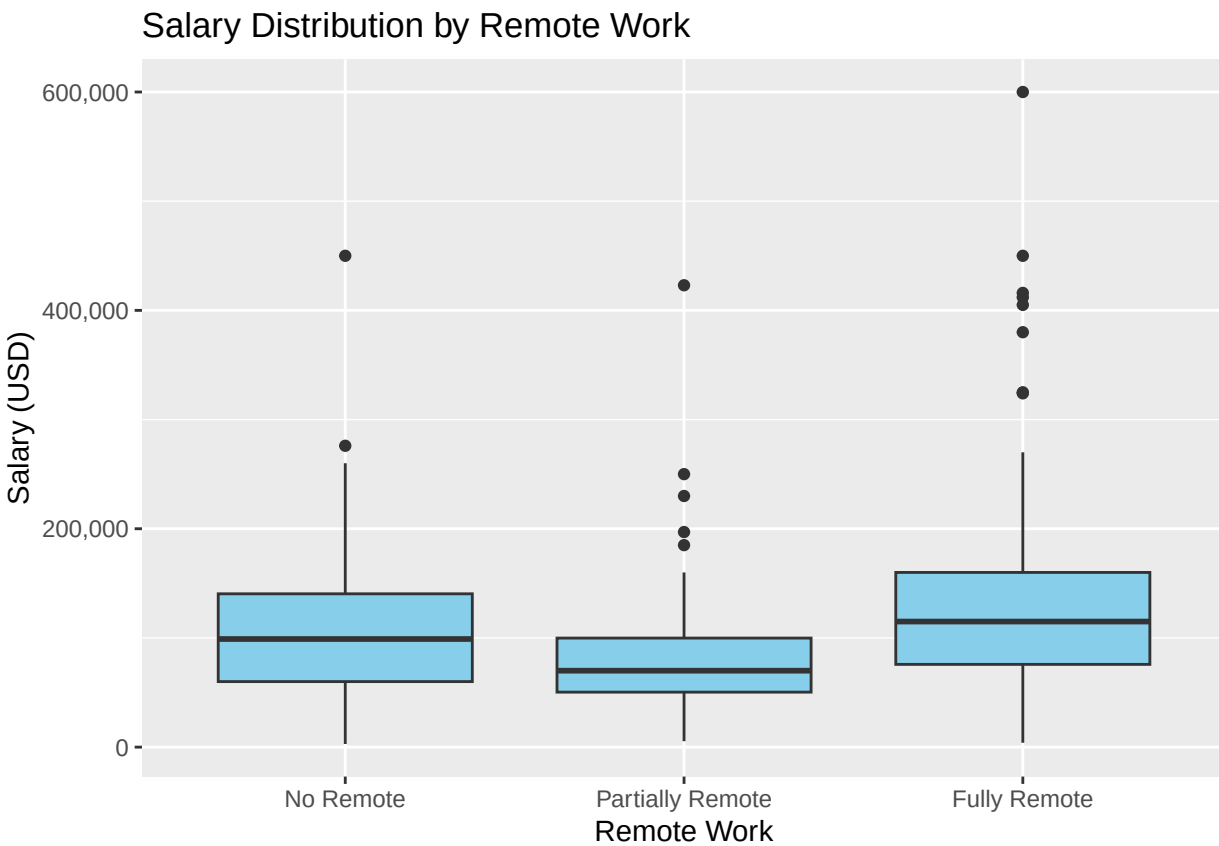

```r
#Salary by Company Size
ggplot(r_project_data_1, aes(x = company_size, y = salary_in_usd)) +
  geom_boxplot(fill = "orange") +
  labs(title = "Salary by Company Size", y = "Salary (USD)", x = "Company Size") +
  scale_y_continuous(labels = label_comma())
```



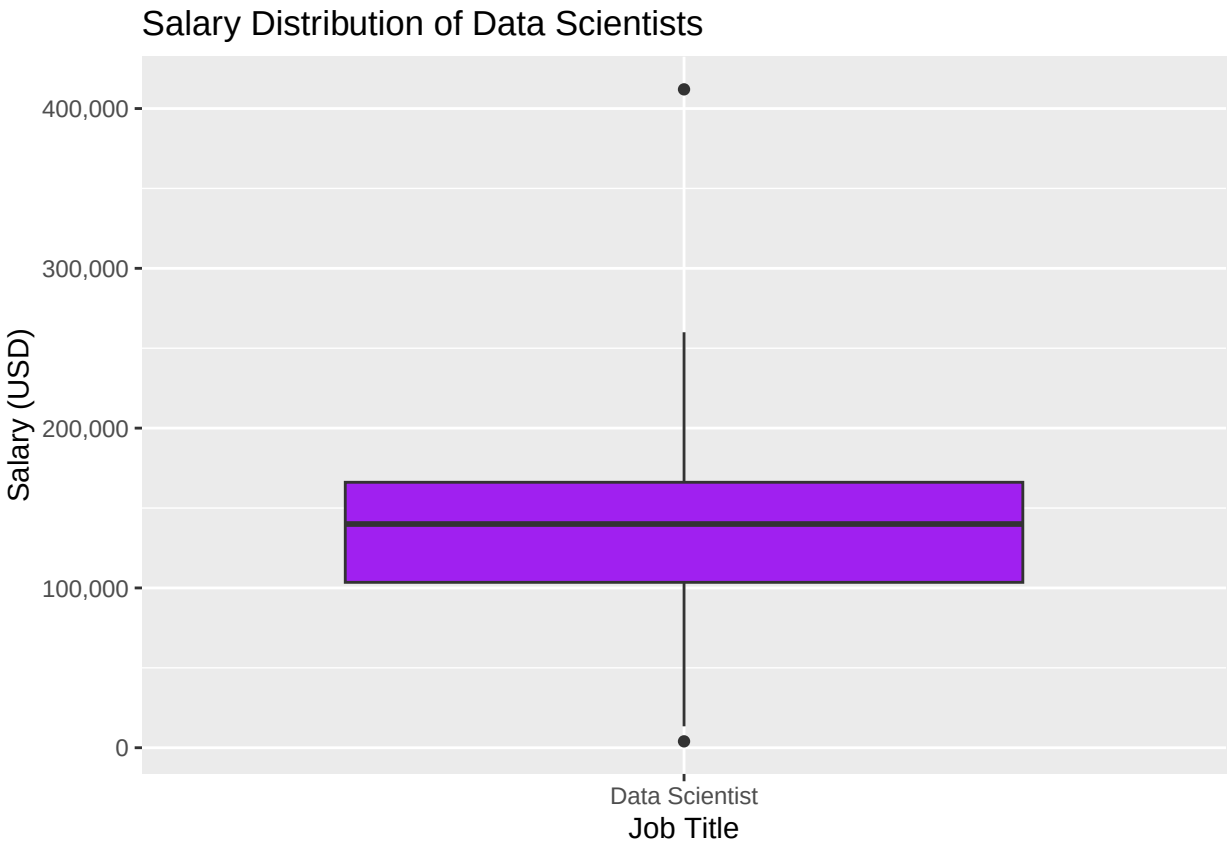

```r
#US vs Offshore Salaries
us_data <- r_project_data_1 %>% filter(employee_residence == "US")
offshore_data <- r_project_data_1 %>% filter(employee_residence != "US")

mean_us_salary <- mean(us_data$salary_in_usd, na.rm = TRUE)
mean_offshore_salary <- mean(offshore_data$salary_in_usd, na.rm = TRUE)

salary_comparison <- data.frame(
  region = c("US", "Offshore"),
  mean_salary = c(mean_us_salary, mean_offshore_salary)
)

ggplot(salary_comparison, aes(x = region, y = mean_salary, fill = region)) +
  geom_bar(stat = "identity") +
  labs(title = "Average Salary: US vs Offshore", y = "Salary (USD)") +
  scale_fill_manual(values = c("US" = "blue", "Offshore" = "red"))
```




```
# Salary by Remote Work
r_project_data_1$remote_ratio <- factor(r_project_data_1$remote_ratio,
                                        levels = c(0, 50, 100),
                                        labels = c("No Remote", "Partially Remote", "Fully Remote"))

r_project_data_1 <- r_project_data_1[!is.na(r_project_data_1$remote_ratio) & !is.na(r_project_data_1$sal

ggplot(r_project_data_1, aes(x = remote_ratio, y = salary_in_usd)) +
  geom_boxplot(fill = "skyblue") +
  labs(title = "Salary Distribution by Remote Work", y = "Salary (USD)", x = "Remote Work") +
  scale_y_continuous(labels = label_comma())
```




```r
# Data Scientist Data
data_scientist_data <- r_project_data_1 |>
  filter(job_title == "Data Scientist", salary_currency == "USD")

head(data_scientist_data)
```

```
##    X work_year experience_level employment_type      job_title salary
## 1 40      2020               MI              FT Data Scientist  45760
## 2 48      2020               MI              FT Data Scientist 105000
## 3 57      2020               MI              FT Data Scientist 118000
## 4 58      2020               SE              FT Data Scientist 120000
## 5 59      2020               MI              FT Data Scientist 138350
## 6 63      2020               SE              FT Data Scientist 412000
##   salary_currency salary_in_usd employee_residence     remote_ratio
## 1             USD         45760                 PH     Fully Remote
## 2             USD        105000                 US     Fully Remote
## 3             USD        118000                 US     Fully Remote
## 4             USD        120000                 US Partially Remote
## 5             USD        138350                 US     Fully Remote
## 6             USD        412000                 US     Fully Remote
##   company_location company_size
## 1               US            S
## 2               US            L
## 3               US            M
## 4               US            L
## 5               US            M
## 6               US            L
```

```r
#Data Scientist Salary Distribution
ggplot(data_scientist_data, aes(x = reorder(job_title, salary_in_usd, FUN = median), y = salary_in_usd)
  geom_boxplot(fill = "purple") +
  labs(title = "Salary Distribution of Data Scientists", y = "Salary (USD)", x = "Job Title") +
  scale_y_continuous(labels = label_comma())
```